

GENERATIVE 6D POSE ESTIMATION VIA CONDITIONAL FLOW MATCHING

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ABSTRACT

Existing methods for instance-level 6D pose estimation typically rely on neural networks that either directly regress the pose in $SE(3)$ or estimate it indirectly via local feature matching. The former struggle with object symmetries, while the latter fail in the absence of distinctive local features. To overcome these limitations, we propose a novel formulation of 6D pose estimation as a conditional flow matching problem in $\mathbb{R}^3$. We introduce Flose, a generative method that infers object poses through a denoising process conditioned on both geometric and appearance-based semantic features. Unlike prior conditional flow matching approaches, which rely solely on geometric guidance, Flose leverages semantic features to mitigate ambiguities caused by object symmetries while incorporating RANSAC-based registration for robustness against outliers. We validate Flose on five datasets from the established BOP benchmark, outperforming prior methods with an average improvement of +4.5 in Average Recall. Project website: <https://tev-fbk.github.io/Flose/>.

Index Terms— 6D pose estimation, conditional flow matching, robot vision

1. INTRODUCTION

While humans interact seamlessly with objects in their environments, this ability remains a fundamental challenge for robotic agents. Object 6D pose estimation aims to bridge this gap by determining an object’s position and orientation in 3D space from sensory images. This capability is fundamental for robotic manipulation [1] and augmented reality [2].

In recent years, the field has transitioned from hand-crafted to learning-based methods, which now achieve state-of-the-art performance [3, 4]. In this work, we focus on learning-based methods that operate under the *model-based instance-level* setting using RGBD data [5, 6, 7, 3]. Model-based methods assume access to a 3D representation of the object, provided

either as a synthetic CAD model [8] or a reconstructed 3D mesh [9, 10, 11, 12]. Instance-level methods rely on direct supervision over a closed set of predefined object instances. These methods can be broadly categorized as *direct* or *indirect*. The former [3, 13, 14, 7] use neural networks to regress the object pose in an end-to-end fashion and are typically trained by minimizing a loss over the $SE(3)$ manifold. They may underperform in the presence of object symmetries and suffer from lower accuracy due to the lack of explicit pixel-to-3D alignment. The latter [4, 15, 16] extract local features to establish object-image correspondences and infer the object pose via robust registration. These methods may fail to establish reliable correspondences in the absence of distinctive local features [17]. Recently, generative modeling has emerged as a promising alternative for 3D point cloud registration by reformulating registration as a progressive denoising process instead of explicit correspondence estimation [18, 19, 20].

In this work, we formulate 6D pose estimation as a conditional flow matching (CFM) [21] problem in $\mathbb{R}^3$ [18] and present Flose (Flow matching for 6D pose estimation), which operates in the model-based instance-level setting. Flose estimates the object pose through a denoising process that learns the displacement field required to register noise samples with the object’s 3D model. Prior CFM-based methods suffer from two main limitations. Because they condition the denoising process solely on geometric guidance, resolving 6D pose ambiguities in symmetric objects where texture provides the only disambiguating cue, can be challenging [18]. Furthermore, because they typically recover the object pose via global alignment (e.g., SVD [22]) between input sensory data and the denoised points, they can be sensitive to displacement outliers [20, 18]. We address these issues by explicitly capturing object appearance and resolving pose ambiguities arising from symmetries through the injection of semantic features extracted using a vision foundation model [23]. We further improve robustness to noisy correspondences and displacement outliers by replacing global alignment with a random sampling approach [24], which identifies a subset of points that are geometrically consistent with a rigid transformation and performs registration exclusively on them. We evaluate Flose on five datasets from the BOP benchmark [9, 8, 10, 11, 12], spanning diverse objects and real-world conditions. We compare Flose against both direct [3, 7] and indirect [4, 15, 25, 6, 16, 26] methods. Compared to the leading method [15] training a

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model per dataset, Flose achieves an improvement of +4.5 Average Recall (AR). Compared to the leading method [3] training a model per object, Flose achieves +1.2 AR while requiring lower training and inference costs. In summary, our contributions are:

- We propose Flose, the first conditional flow matching formulation for instance-level object 6D pose estimation.
- We integrate semantic features from vision foundation models to disambiguate object symmetries and occlusions.
- We employ robust RANSAC-based registration to effectively filter outliers produced during the denoising process.

2. RELATED WORK

Instance-level methods for 6D pose estimation regress the poses of a closed set of predefined object instances and can be categorized into direct and indirect approaches. Indirect methods first establish image-object correspondences and then recover the pose using Perspective-n-Point (PnP) or RANSAC algorithms. SurfEmb [25] learns dense image-object correspondences through surface embeddings. ZebraPose [26] introduces a hierarchical encoding of the visible object surface, while HccePose(BF) [4] extends this representation to back-facing surfaces, enabling denser correspondences. Other indirect approaches refine coarse pose predictions by enforcing consistency across objects or views [27, 7, 15]. These methods are constrained by the quality of the estimated correspondences. In contrast, direct methods regress an object’s 6D pose in an end-to-end manner by minimizing a loss over the SE(3) manifold. GDR-Net [13] regresses poses using a differentiable patch-based PnP, while GDRNPP [3] achieves state-of-the-art performance by incorporating stronger backbones and domain randomization. These methods struggle with supervision ambiguity when handling symmetric objects that admit multiple valid poses despite having a single ground-truth pose during training. By formulating Flose as conditional flow matching in $\mathbb{R}^3$, we mitigate sensitivity to local correspondence outliers through dense displacement regression and alleviate the rotational ambiguities induced by SE(3)-based losses.

Generative methods for 6D pose estimation use diffusion [28] or flow matching [21] to model a distribution of valid 6D poses instead of regressing a single prediction [29, 30, 31]. They primarily model pose ambiguity from occlusions (e.g., a mug with an occluded handle) or shape symmetries (e.g., a cereal box, a tuna can) by generating multiple pose hypotheses via stochastic sampling. DiffusionNOCs [29] learns to denoise NOCs [32] coordinates to recover 6D poses. GenPose [30] frames pose estimation as a conditional diffusion process, while SE(3)-PoseFlow [31] applies flow matching on the SE(3) manifold. These methods operate in the category-level setting, relying on weaker supervision signals than instance-level approaches. Thus, a direct quantitative comparison with Flose is not applicable, as we target accurate pose estimation where explicit 3D models of object instances are available.

Flow matching for 3D tasks has recently proven effective

for 3D shape assembly [21, 18, 19] and point cloud registration [20] by modeling alignment as a generative denoising process. GARF [19] applies flow matching over the SE(3) manifold to generate relative poses, whereas RPF [18] operates directly in $\mathbb{R}^3$ to learn point cloud denoising without enforcing rigidity constraints. RAP [20] further extends RPF by incorporating rigidity constraints into the Euler integration scheme during inference. Similarly to RPF and RAP, we employ flow matching in $\mathbb{R}^3$. However, we address the distinct problem of object 6D pose estimation. Unlike 3D registration, which typically aligns two point cloud fragments acquired from the same sensor, 6D pose estimation requires aligning a clean, complete CAD model with a noisy, partial real-world observation affected by clutter and occlusions.

3. METHOD

Problem formulation. Given the 3D model of a *query object* and an RGBD image $\mathbf{I}$ of a scene containing it, our goal is to recover the 6D pose of the object with respect to the camera reference frame. We define the *target object* as the partial observation of the query object visible in $\mathbf{I}$. Let $\mathcal{Q} \in \mathbb{R}^{N \times 3}$ be the point cloud of the query object, and let $\mathcal{T} \in \mathbb{R}^{N \times 3}$ be the point cloud obtained by lifting the pixels corresponding to the target object into 3D space using the camera intrinsic parameters. For notational simplicity, we assume the two point clouds have the same number of points N . Our objective is to estimate the six degrees-of-freedom rigid transformation ($\hat{\mathbf{R}} \in \text{SO}(3), \hat{\mathbf{t}} \in \mathbb{R}^3$) that aligns $\mathcal{Q}$ with $\mathcal{T}$, such that $\mathcal{Q}^r = \hat{\mathbf{R}}\mathcal{Q} + \hat{\mathbf{t}} \approx \mathcal{T}$. For the alignment process, we further define $\mathcal{T}^r$ as the target point cloud transformed into the canonical frame of the query object using the ground-truth transformation.

Overview. We design a three-stage pipeline as in Fig. 1. First, we extract point-wise features $\mathbf{F}^{\mathcal{Q}}, \mathbf{F}^{\mathcal{T}}$ from $\mathcal{Q}, \mathcal{T}$ by fusing overlap-aware geometric features and appearance-aware semantic features from encoders Φ_{Θ} and Γ . Next, we deform $\mathcal{T}$ into $\hat{\mathcal{T}} \approx \mathcal{T}^r$ using a flow matching model Ψ_{Ω} conditioned on these features. Finally, we estimate the rigid transformation ($\mathbf{R}, \mathbf{t}$) aligning $\mathcal{T}$ with $\hat{\mathcal{T}}$ by solving a Procrustes problem.

Feature encoding extracts point-level features $\mathbf{F}$ by combining overlap-aware features $\mathbf{O}$ and semantic features $\mathbf{S}$ via point-wise addition. The overlap-aware encoder Φ_{Θ} is a parametric neural network with learnable parameters Θ . It takes 3D points and their normals (estimated via PCA over local neighborhood) as input and predicts which points belong to the overlapping region between $\mathcal{Q}$ and $\mathcal{T}^r$. Applying Φ_{Θ} to $\mathcal{Q}$ and $\mathcal{T}$ produce binary classification logits $\mathbf{L}^{\mathcal{Q}}, \mathbf{L}^{\mathcal{T}} \in \mathbb{R}^{N \times 2}$. During training, the parameters Θ are optimized via binary cross-entropy loss between predicted logits $\mathbf{L}^{\mathcal{Q}}, \mathbf{L}^{\mathcal{T}}$ and ground-truth binary labels $\mathbf{y}^{\mathcal{Q}}, \mathbf{y}^{\mathcal{T}} \in [0, 1]^N$, indicating the points of $\mathcal{Q}$ and $\mathcal{T}$ that belong to the overlap region. $\mathbf{y}^{\mathcal{Q}}, \mathbf{y}^{\mathcal{T}}$ are constructed via nearest-neighbor search in Euclidean space: during training, $\mathcal{T}$ is first transformed to the query frame using the ground-truth transformation to obtain $\mathcal{T}^r$, then pairwise distances between

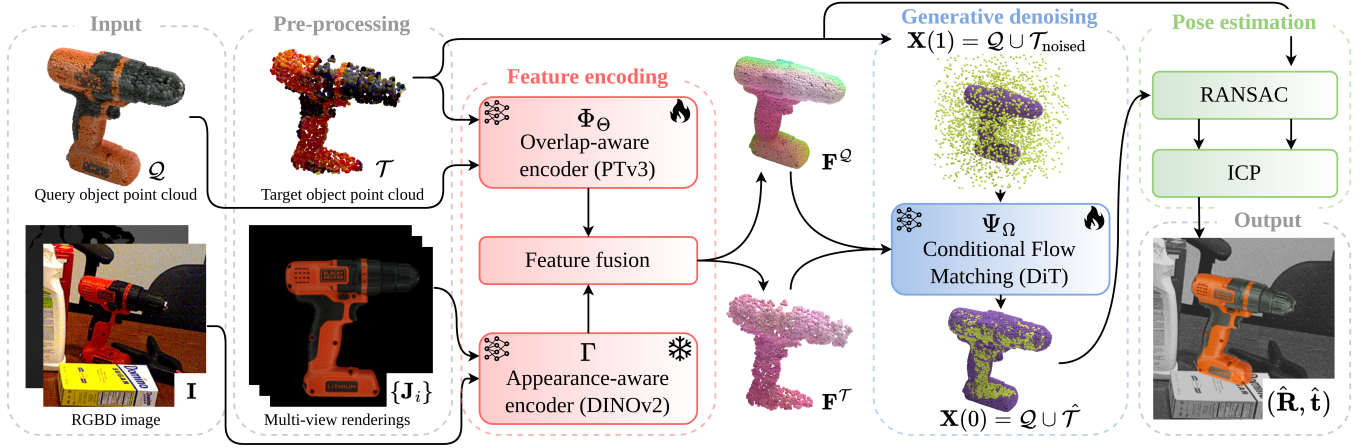


Fig. 1. Overview of FloSe. Given the query object point cloud Q and an RGBD image I as input (left), FloSe estimates the 6D pose $(\hat{\mathbf{R}}, \hat{\mathbf{t}})$ (bottom right) through three stages: feature encoding (red), generative denoising (blue) and pose estimation (green). *Feature encoding*: An overlap-aware encoder Φ_Θ and appearance-aware encoder Γ produce per-point descriptors that are fused to produce $\mathbf{F}^Q, \mathbf{F}^T$. Colors encode feature similarity: Corresponding regions share similar colors (overlap-awareness), while semantically distinct parts differ (appearance-awareness). *Generative denoising*: A generative network Ψ_Ω , conditioned on $\mathbf{F}^Q, \mathbf{F}^T$, learns a displacement field that iteratively denoise the Gaussian noised $\mathbf{X}(1)$ into an aligned position $\mathbf{X}(0)$. *Pose estimation*: The 6D pose $(\hat{\mathbf{R}}, \hat{\mathbf{t}})$ is recovered via RANSAC-based Kabsch solver followed by ICP refinement.

Q and T^r are computed, and points with distances smaller than 1% of the object’s diameter are labeled as overlapping. During inference, the classification head of Φ_Θ is discarded, and only the hidden features $\mathbf{O}^Q, \mathbf{O}^T \in \mathbb{R}^{N \times F}$ are retained, yielding overlap-aware features. However, relying solely on the geometric cues is insufficient for resolving ambiguities in objects with symmetries and sparse geometric details. We incorporate robust appearance information by using a semantic encoder which associates pixel-level features from a frozen vision foundation model (VFM) to the 3D points of Q and T . For the target object, the VFM is applied to a regular crop of I centered on its barycenter and uses the known pixel-to-point correspondence to associate the pixel-level features with the points of T , producing point-level features $\mathbf{S}^T \in \mathbb{R}^{N \times G}$, where G is the dimensionality of the VFM features. For the query object, the encoder generates multi-view renderings $\{J_i\}$ of its 3D model, applies the VFM to the synthetic images, establishes pixel-to-point correspondences using the synthetic camera intrinsics, and associates the pixel-level features to the corresponding 3D points. If multiple pixels map to the same 3D point, their features are fused via average pooling. This process is performed once per object during pre-processing; at inference, the query features $\mathbf{S}^Q \in \mathbb{R}^{N \times G}$ are loaded from memory. The dimensionality of $\mathbf{S}^Q$ and $\mathbf{S}^T$ is reduced to F via PCA to match the overlap-aware features. Specifically, we compute the PCA basis only on $\mathbf{S}^Q$ and apply this projection to both $\mathbf{S}^Q$ and all instances of $\mathbf{S}^T$. Finally, the output features are defined by combining overlap-aware and semantic features via point-wise addition, i.e., $\mathbf{F} = \text{norm}(\text{norm}(\mathbf{O}) + \text{norm}(\mathbf{S})) \in \mathbb{R}^{N \times F}$, where $\text{norm}(\cdot)$ denotes the operator that normalizes each point feature to have unit L_2 norm, for balanced representation.

Conditional flow matching [18] transforms a source point cloud $\mathbf{X}(0)$ into a target point cloud $\mathbf{X}(1)$ via linear interpolation, $\mathbf{X}(t) = (1 - t)\mathbf{X}(0) + t\mathbf{X}(1)$, with $t \in [0, 1]$. In our setting, $\mathbf{X}(0) = Q \cup T^r \in \mathbb{R}^{2N \times 3}$, while $\mathbf{X}(1) \in \mathbb{R}^{2N \times 3}$ is obtained by uniformly sampling Gaussian noise in 3D space. The neural network $\Psi_\Omega(t, \mathbf{X}(t)|\mathbf{C})$ learns to reverse this process by predicting a vector field $\mathbf{V} \approx \mathbf{X}(0) - \mathbf{X}(t) \in \mathbb{R}^{2N \times 3}$ that transforms $\mathbf{X}(t)$ into $\mathbf{X}(0)$. Ψ_Ω is conditioned on overlap-aware and semantic features $\mathbf{C}$. The parameters Ω are optimized using the conditional flow matching loss [18]. At inference time, we perform K uniform Euler steps: $\hat{\mathbf{X}}(t - \Delta t) = \hat{\mathbf{X}}(t) - \Psi_\Omega(t, \mathbf{X}(t)|\mathbf{C})\Delta t$, where $\Delta t = 1/K$. After K iterations, the resulting $\hat{\mathbf{X}}(0) = Q \cup \hat{T}$ approximates $\mathbf{X}(0) = Q \cup T^r$. In practice, we replace the components of $\mathbf{V}$ corresponding to the anchor Q with a null field and apply the resulting velocity field to T , obtaining the deformed point cloud $\hat{T} = T + \mathbf{V} \approx T^r$. Unlike [18], which conditions Ψ_Ω solely on geometric encoding, we enrich the conditioning signal $\mathbf{C} \in \mathbb{R}^{N \times (F+H)}$ with overlap-aware and semantic point-wise features that jointly capture geometric structure and object appearance. Specifically, $\mathbf{C} = [\mathbf{F} \in \mathbb{R}^{N \times F} | \mathbf{P} \in \mathbb{R}^{N \times H}]$, where $\mathbf{F}$ is extracted by the feature encoder, $\mathbf{P}$ is obtained via the positional encoder Ξ , and $|$ denotes concatenation along the last dimension. For each point, Ξ encodes a 10-dimensional vector consisting of coordinates, normals, coordinates of the corresponding point in the noisy initialization, and a binary indicator specifying whether the point belongs to Q or T . In practice, Ξ applies sinusoidal embeddings with logarithmically spaced frequencies to each geometric attribute independently [33]. The resulting embeddings are concatenated with learned point-wise features, and a linear projection is applied to obtain the final positional

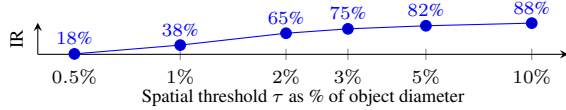


Fig. 2. Sensitivity of the inlier ratio (IR) to the spatial threshold τ . IR is the percentage of points in $\hat{\mathcal{T}}$ whose distance to the corresponding points in $\mathcal{T}^r$ is smaller than τ . At strict thresholds, the majority ($> 80\%$) of correspondences are outliers.

encoding $\mathbf{P}$ used to condition the flow model Ψ_Ω .

Pose estimation. Ψ_Ω predicts $\hat{\mathcal{T}}$, a non-linear deformation of $\mathcal{T}$ that approximates its rigidly aligned counterpart $\mathcal{T}^r$. Since flow fields are predicted per point without explicit rigidity constraints, the resulting $\hat{\mathcal{T}}$ is not guaranteed to be a perfect rigid transformation of $\mathcal{T}$. Fig. 2 quantifies this effect: at a strict threshold $\tau = 0.5\%$ of the object’s diameter, only 18% of correspondences are inliers, and even at $\tau = 2\%$ nearly 35% of points remain outliers on all instances of LM-O. Global alignment via SVD over all correspondences is therefore dominated by outliers, leading to inaccurate pose estimates. We therefore employ RANSAC, where each iteration samples a minimal set of three correspondences to solve the orthogonal Procrustes problem using the Kabsch algorithm. The candidate transformation with the highest number of inliers is selected as the initial estimate $(\mathbf{R}, \mathbf{t})$, representing the transformation from the target to the query frame. We refine this pose using ICP. Finally, we invert this transformation to obtain the standard 6D pose of the query object in the camera frame.

4. EXPERIMENTS

Implementation details. During pre-processing, the target object from input image is segmented using ZebraPose [26] masks publicly available on the BOP leaderboard. In the feature extraction stage, Φ_Θ is based on PointTransformerV3 [34] and produces overlap-aware features of size $F = 64$, while Γ is based on a frozen DINOv2-Giant [23], yielding appearance-aware features of dimension $G = 1536$, reduced to F via PCA. These features are then fused via point-wise summation. In the flow matching stage, the overlap and semantic features are concatenated with position-aware features of size $H = 210$, derived from the positional encoding of object coordinates, normals, and noisy point positions. The resulting features of size $F + H = 274$ are fed to Ψ_Ω , which is based on Diffusion Transformer [35]. During training, Θ and Ω are initialized with RPF [18] checkpoints and finetuned on the training split of each dataset in two separate phases for 200 and 100 epochs, respectively. Training is distributed across 12 NVIDIA A100 GPUs with an effective batch size of 384. During post-processing, RANSAC registration is applied for 1k iterations with an inlier threshold of 1 cm, followed by ICP refinement for 3k iterations with a distance threshold of 1 cm. **Experimental validation.** We validate Flore on five datasets from the BOP benchmark [9, 8, 10, 11, 12], featuring diverse

Table 1. Comparison with RGBD methods in terms of AR. The top block (rows 1-5) trains a separate model for each object, while the bottom block (rows 6-11) trains a single model (S.M.) per dataset. Row 12 quantifies the absolute improvement of Flore over the strongest S.M. competitor.

Method	S.M.	LM-O	T-LESS	TUD-L	IC-BIN	YCB-V	Avg
1 Pix2Pose [16]		58.8	51.2	82.0	39.0	78.8	62.0
2 ZebraPose [26]		75.2	72.7	94.8	65.2	86.6	78.9
3 GDRNPP (BOP22) [3]		77.5	87.4	96.6	72.2	92.1	85.2
4 HcPose(BF) [4]		80.5	87.9	94.4	72.4	91.1	85.3
5 GDRNPP (BOP23) [3]		79.4	91.4	96.4	73.7	92.8	86.7
6 Koenig-Hybrid [6]	✓	63.1	65.5	92.0	43.0	70.1	66.7
7 CosyPose [27]	✓	71.4	70.1	93.9	64.7	86.1	77.2
8 SurfEmb [25]	✓	75.8	83.3	93.3	65.6	82.4	80.1
9 CIR [7]	✓	73.4	77.6	96.8	67.6	89.3	81.0
10 PFA [15]	✓	79.7	85.0	96.0	67.6	88.8	83.4
11 Flore (ours)	✓	86.1	86.9	98.8	74.8	92.8	87.9
12 Improv. over row 10		+6.4	+1.9	+2.8	+7.2	+4.0	+4.5

object types (everyday vs. industrial), varying appearance (textured vs. texture-less), and complex symmetries, as well as challenging environmental conditions such as heavy occlusion and changing lighting. We compare Flore against ten peer-reviewed competitors, categorized into two protocols: (i) *per-object* methods that train a dedicated model for each object instance, such as ZebraPose [26], GDRNPP [3], and HcPose (BF) [4]; and (ii) *per-dataset* methods that train a single model for all objects within a dataset, such as SurfEmb [25], CIR [7], and PFA [15]. We adhere to the latter setting. Following the standard BOP protocol, we evaluate pose accuracy using the Average Recall (AR) metric [36].

Quantitative results. Tab. 1 details pose accuracy for various instance-level RGBD methods. Competitor results are reported from [3, 4]. The bottom block (rows 6-11) lists per-dataset methods that are directly comparable to Flore, as they train a single model per dataset (✓ in S.M. column, 5 models total). Flore achieves an average improvement of +4.5 AR over PFA [15], the strongest competitor in this category. The performance gains are consistent across all datasets (row 12). The top block (rows 1-5) shows per-object methods that train one model per object (54 models total). Even under this more challenging comparison, Flore outperforms the current state-of-the-art GDRNPP [3] by +1.2 AR on average. We further observe that the gain in AR is higher for symmetric objects: on LM-O, the gain for the symmetric Eggbox and Glue bottle is +3.95 AR, compared to +2.12 AR for non-symmetric objects, highlighting the effectiveness of semantic features in resolving pose ambiguities. Flore achieves these results while relying on less specialized models (the same model is used for all object instances of a dataset) and requiring less compute and memory resources (54/5 $\approx 11\times$ fewer models to train and load).

Qualitative results. Fig. 3 shows qualitative comparisons on LM-O and YCB-V between Flore and an RPF-based [18] baseline finetuned on the same data. Each image shows a rendering of the object’s 3D model in the predicted pose, overlaid on a grayscale version of the input image. Prediction accuracy is



Fig. 3. Flore (center) vs. an RPF-based [18] baseline adapted for pose estimation (right). Flore predicts more accurate poses under severe occlusions (rows 1-2) and handles symmetry ambiguities where pure geometric methods fail (rows 3-4).

assessed by the alignment between the colored rendering and the visible portion of the object in the grayscale background image. Rows 1-2 show a red toy ape and a white watering can under severe occlusions caused by hand-object interaction and clutter. Rows 3-4 show objects with varying degrees of symmetry: a red mug with uniform texture, and a blue Master Chef can with a cylindrical shape (infinite rotational symmetry) and rich texture (brand name and logo). Flore outperforms RPF across all scenarios, showing robustness to real-world noise.

Ablation and hyperparameter study on LM-O using ZebraPose [26] segmentation masks. Fig. 4(a) analyzes the impact of feature conditioning. The top plot shows that fusing appearance-based [23] and overlap-aware [18] features surpasses either modality in isolation, yielding gains of +15.0 and +2.6 AR, respectively. The bottom plot shows that this improvement stems from consistently higher inlier ratios (IR) compared to the geometry-only baseline. The gain is most pronounced at strict thresholds, confirming that semantic information provides discriminative cues necessary for establishing accurate correspondences. Fig. 4(b) compares different pose estimation strategies, showing that RANSAC outperforms SVD by robustly filtering outliers generated by the denoising

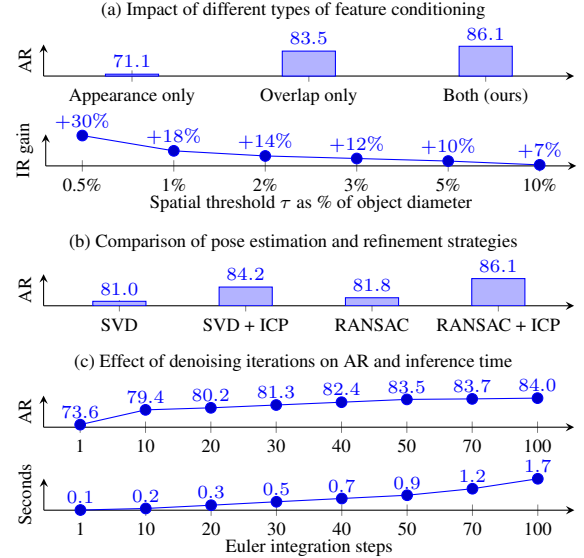


Fig. 4. Ablation and hyperparameter study on LM-O: (a) Impact of the conditioning features on the flow matching process, measured in terms of AR (top) and IR gain relative to a baseline using only overlap-aware features (bottom); (b) Comparison of pose solvers (SVD vs. RANSAC) and the effect of ICP-based refinement; (c) Pose accuracy (top) and inference time (bottom) as a function of Euler integration steps.

process. Furthermore, applying ICP-based geometric refinement improves accuracy by an additional +4.3 AR, effectively correcting residual alignment errors from the coarse pose estimate. Fig. 4(c) analyzes the impact of the number of Euler integration steps. AR improves with the number of iterations and saturates at higher values (top). The average inference time per image scales linearly from 0.1 to 1.7 seconds (bottom). This parameter enables an accuracy-efficiency trade-off.

5. CONCLUSIONS

We presented Flore, the first instance-level 6D pose estimation method formulated as conditional flow matching in $\mathbb{R}^3$. Flore conditions the generative denoising process on a fusion of overlap-aware and semantic features, and employs robust RANSAC-based registration. Experiments demonstrate that Flore outperforms state-of-the-art competitors, achieving superior robustness to symmetries and occlusions while requiring less training and smaller inference costs. Our iterative formulation enables explicit control over the accuracy-efficiency trade-off by adjusting the number of Euler integration steps.

Limitations and future work: (1) Our current pipeline relies on a two-stage training process, which could be simplified using off-the-shelf geometric and semantic descriptors. (2) The iterative denoising process of flow matching hinders time-critical applications; enabling single-step denoising is a promising direction for future work. (3) Flore currently depends on object-level segmentation; extending it to operate directly at the scene level is an area for future work.

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